claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-af76cd4b201a7aa53.jsonl`
- SHA-256: `c8b2c41703480bf8b11787b6bcbbba7eb0bfd624bf44666ea551fa4fdcc6f887`
- Source modified: `2026-05-31T04:23:07+00:00`
- Imported at: `2026-07-05T16:48:21+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:21:06.034Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On the olmOCR-Bench evaluation, olmOCR 2 scores 84.9 on tables and 83.7 on multi-column layouts, with the paper attributing its largest gains specifically to table parsing and multi-column layouts ? the exact document features muneem needs for dense, multi-column bank statements."

Source: https://arxiv.org/html/2510.19817v1 (primary)
Supporting quote: "largest improvements in math formula conversion, table parsing, and multi-column layouts"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:09.442Z)

I'll verify this claim about olmOCR 2's benchmark scores. Let me fetch the primary source first.

3. assistant (2026-05-31T04:21:12.978Z)

I'll fetch the primary source (the olmOCR 2 arXiv paper) to verify the specific table and multi-column scores claimed.

4. user (2026-05-31T04:21:17.742Z)

olmOCR 2 Benchmark Results

Based on Table 3 in the document, here are the requested evaluation results:

(1) Tables category score:
olmOCR 2 achieves **84.9** on the tables category.

(2) Multi-column layouts category score:
olmOCR 2 achieves **83.7** on the multi-column category.

(3) Statement on largest improvements:

The abstract explicitly states: "with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions."

This sentence directly identifies which categories demonstrated the greatest gains for olmOCR 2 relative to the initial olmOCR release.

5. user (2026-05-31T04:21:21.043Z)

Web search results for query: "olmOCR 2 olmOCR-Bench tables multi-column score 84.9 83.7"

Links: [{"title":"olmOCR 2: Unit Test Rewards for Document OCR | alphaXiv","url":"https://www.alphaxiv.org/overview/2510.19817v1"},{"title":"olmOCR 2: Advanced Open-Source OCR","url":"https://www.emergentmind.com/topics/olmocr-2"},{"title":"OlmOCR-Bench: OCR Evaluation Benchmark","url":"https://www.emergentmind.com/topics/olmocr-bench"},{"title":"olmOCR 2: Unit test rewards for document OCR | Ai2","url":"https://allenai.org/blog/olmocr-2"},{"title":"olmOCR 2: Unit Test Rewards for Document OCR","url":"https://arxiv.org/pdf/2510.19817"},{"title":"OlmOCR-Bench Review: Insights and Pitfalls | LlamaIndex","url":"https://www.llamaindex.ai/blog/olmocr-bench-review-insights-and-pitfalls-on-an-ocr-benchmark"},{"title":"olmocr/olmocr/bench at main · allenai/olmocr","url":"https://github.com/allenai/olmocr/tree/main/olmocr/bench"},{"title":"GitHub - allenai/olmocr: Toolkit for linearizing PDFs for LLM datasets/training · GitHub","url":"https://github.com/allenai/olmocr"},{"title":"Papers Explained 480: olmOCR 2 | by Ritvik Rastogi | Medium","url":"https://ritvik19.medium.com/papers-explained-480-olmocr-2-33fbf89757eb"}]

Based on the search results, I found information about olmOCR 2 and its performance on the olmOCR-Bench benchmark related to the scores you mentioned.

olmOCR 2 achieved a score of 84.9 for math formula conversion and achieved improvements in multi-column layouts. The search results also indicate that table parsing improved to 47.7, though this is different from the 83.7 score you mentioned.

More broadly, olmOCR 2 achieves an overall score of 82.4 ± 1.1 on the olmOCR-Bench benchmark, representing a substantial +14.2 point improvement over its predecessor.

The 84.9 score appears to be specifically for math formula conversion performance, which was one of the areas with the largest improvements in the olmOCR 2 release. If you're looking for more specific information about the 83.7 score, I may need to search for additional details or clarify what metric that refers to.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:24.601Z)

There's a discrepancy. WebFetch (which reads the actual paper HTML, Table 3) confirms tables=84.9 and multi-column=83.7. But the WebSearch summary claims 84.9 is the math formula score and tables is 47.7. This is a conflict I must resolve by reading the actual benchmark table. Let me fetch the alphaXiv overview and the Medium paper explanation, which often reproduce the full benchmark table verbatim.

7. user (2026-05-31T04:21:27.415Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

8. user (2026-05-31T04:21:31.990Z)

I cannot find the olmOCR-Bench benchmark results table with the specific per-category scores you're requesting in the provided content. While the article mentions "OCR model performance comparison" with a reference to an image, the actual numerical results table is not included in the text.

The article does not contain quoted sentences about which categories showed the largest improvements or specific scores for "Tables" or "Multi-column" categories. To obtain this information, you would need to access the full paper (arxiv 2510.19817) or view the referenced benchmark results table directly.

9. assistant (2026-05-31T04:21:35.417Z)

The Medium article doesn't have the table inline, and alphaXiv returned 403. Let me go directly to the arXiv PDF and also re-query the HTML for the full Table 3 with all column headers, so I can confirm the column mapping (whether 84.9/83.7 are really tables/multi-column, vs the WebSearch claim that 84.9 is math).

10. user (2026-05-31T04:21:42.781Z)

I apologize, but I cannot locate Table 3 (olmOCR-Bench results) in the PDF content provided. The document appears to be truncated, and while I can see references to tables and figures throughout the metadata and structure, the actual table data with column headers and olmOCR 2 scores is not included in the accessible text.

To properly answer your question about:
- All column headers in order
- The "Tables" column score for olmOCR 2
- The "Multi column" score for olmOCR 2
- The abstract quote about largest improvements

I would need access to the complete rendered table content, which isn't visible in this PDF structure dump.

I recommend accessing the full paper directly at https://arxiv.org/abs/2510.19817 to view Table 3 with complete scoring data.

[Binary content (application/pdf, 1.8MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\tool-results\webfetch-1780201298991-vzygw7.pdf]

11. user (2026-05-31T04:21:43.565Z)

Table 3 Column Headers and olmOCR 2 Values

Column headers in order:
ArXiv, Old scans math, Tables, Old scans, Headers & footers, Multi column, Long tiny text, Base, Overall

olmOCR 2 row values mapped to columns:
- ArXiv: 83.0
- Old scans math: 82.3
- Tables: 84.9
- Old scans: 47.7
- Headers & footers: 96.1

- Multi column: 83.7
- Long tiny text: 81.9
- Base: 99.7
- Overall: 82.4±1.1

To answer your specific questions:
Yes, the "Tables" cell shows 84.9. Yes, the "Multi column" cell displays 83.7. The mathematical formula score is in the "Old scans math" column: 82.3.

12. assistant (2026-05-31T04:21:47.784Z)

The PDF was saved locally. Let me verify the column mapping directly against the saved PDF to be certain, since the WebSearch summary claimed 84.9 was the math score (the "Old scans" column is 47.7, which explains where the WebSearch model got confused).

13. user (2026-05-31T04:21:48.576Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

14. assistant (2026-05-31T04:21:52.121Z)

PDF rendering isn't available. Let me extract the text layer directly with Python (pypdf is in this project's deps) to find Table 3 and the abstract sentence verbatim.

15. user (2026-05-31T04:21:56.225Z)

```
Traceback (most recent call last):
  File "<string>", line 2, in <module>
    import pypdf
ModuleNotFoundError: No module named 'pypdf'
```

16. user (2026-05-31T04:22:00.892Z)

```
FOUND: .venv/Scripts/python.exe
/usr/bin/bash: line 1: uv: command not found
```

17. user (2026-05-31T04:22:09.183Z)

```
=== ABSTRACT ===
'ewards are a diverse set of binary unit tests.\nTo scale unit test creation, we develop a
pipeline for generating synthetic documents with diverse\nand challenging layouts, known ground-
truth HTML source code, and extracted test cases. We show\nthat RL training on these test cases
results in state-of-the-art performance onolmOCR-Bench, our\nEnglish-language OCR benchmark,
with the largest improvements in math formula conversion, table\nparsing, and multi-column
layouts compared to previous versions. We release our model, data and\ncode under permissive
open licenses.\nCodeallenai/olmocr\nDemoolmocr.allenai.org\nDataolmOCR-mix-1025 olmOCR-syn'

=== ALL olmOCR 2 ROWS ===
'olmOCR 2\nUnit Test Rewards for Document OCR\nJake Poznanski Luca Soldaini Kyle Lo\nAllen
Institute for AI{jakep|lucas|kylel}@allenai.org\nAbstract\nWe presentolmOCR 2, the latest in our
family of powerful OCR systems for converting di'
---
'olmOCR 2, the latest in our family of powerful OCR systems for converting digitized\nprint
documents, like PDFs, into clean, naturally ordered plain text.olmOCR 2is powered
by\nolmOCR-2-7B-1025, a specialized, 7B vision language mod'
---
'olmOCR-2-7B-1025, a specialized, 7B vision language model (VLM) trained using
reinforcement\nlearning with verifiable rewards (RLVR), where our rewards are a diverse set of
binary unit tests.\nTo scale unit test creation, we develop'
---
'olmOCR-2-7B-1025 olmOCR-2-7B-1025-FP8\n1 Introduction\nSince our initial release
ofolmOCR(Poznanski et al., 2025) in February 2025, we?ve seen an explosion of\nprogress in
advancing the state-of-the-art in optical character recogniti'
---
```

```
'olmOCR-2-7B-1025-FP8\n1 Introduction\nSince our initial release ofolmOCR(Poznanski et al.,
2025) in February 2025, we?ve seen an explosion of\nprogress in advancing the state-of-the-art
in optical character recognition (OCR). In this'
---
'olmOCR 2?a state-of-the-art OCR system for extracting and linearizing content\nfrom digitized
print documents like PDFs.olmOCR 2is powered byolmOCR-2-7B-1025, an OCR-specialized\nVLM trained
using reinforcement learning with verifia'
---
'olmOCR-2-7B-1025, an OCR-specialized\nVLM trained using reinforcement learning with verifiable
rewards (RLVR) (Lambert et al., 2024). Our training\nrecipe involves two parts:\n1. We develop a
synthetic document pipeline that can take'
---
'olmOCR2-synthmix-1025, our final data mix consists of 2,186 PDF pages. In total, across these
PDF pages,\nwe create 30,381 test cases.\nAlongside olmOCR2-synthmix-1025, we use a refreshed
mix for supervised fine-tuning,olmOCR-mix-10'
---
'olmOCR2-synthmix-1025, we use a refreshed mix for supervised fine-tuning,olmOCR-mix-1025.\nThe
dataset contains 267,962 pages from over 100,000 PDFs sampled from diverse sources, including
9,828\npages from national archives. Compar'
---
'olmOCR2-synthmix-1025 using an 8xH100 GPU\nnode. For each document, 28 completions are
generated. Each completion gets scored using the standard\nolmOCR-Benchscoring rules, where each
test case is either a pass or fail, and the rewa'
---
'olmOCR 2?s RLVR training. Given a generated HTMl page and its unit tests (left),\nwe can easily
score a generated Markdown page (right) according to these unit tests. Each test contributes a
binary\nreward which is aggregated at a p'
---
'olmOCR 2,\nevaluated on the latest version ofolmOCR-Bench. We also include a number of powerful
OCR baselines,\nincluding the latest versions of actively developed open OCR projects like
Marker, MinerU, and PaddleOCR,\nas well as som'
---
'olmOCR 2.Finally, our latest releaseolmOCR 2demonstrates a significant improvement in
benchmark\nperformance. Our best model, reported here, is the result of:\n1. A single epoch of
SFT training onolmOCR-mix-1025,\n2. A single epoch o'
---
'olmOCR2-synthmix-1025,\n3. Repeating the RL training for six random seeds and averaging (or
?souping?) the checkpoints. We used\nimportance sampling at both the token level (3 runs) and
the sequence level (3 runs); more details on\nt'
---
'olmOCR 2, a state-of-the-art OCR system powered by an OCR-specialized VLM trained\nusing
reinforcement learning with verifiable rewards. We define these rewards using binary unit tests
and scale\nthe generation of these tests throug'
---
'olmOCR 2. We thank Haydn Jones, Charitarth Chugh,\nand Vik Paruchuri for their contributions to
our open source repo. We thank the Qwen team for releasing\nopen VLM models that have
accelerated this exciting line of work. We thank t'
---
```

18. assistant (2026-05-31T04:22:12.937Z)

```
I've confirmed the abstract sentence verbatim. Now let me extract the actual Table 3 numeric row
to definitively confirm the 84.9 (tables) and 83.7 (multi-column) mapping, resolving the
WebSearch discrepancy.
```

19. user (2026-05-31T04:22:19.289Z)

```
=== region near 'Multi' ===
ts.
ArXiv
Old
scans
math
Tables Old
scans
```

```
Headers
& footers
Multi
column
Long
tiny
text
Base Overall
olmOCR-mix-022578.6 79.9 72.9 43.9 95.1 77.3 81.2 98.9 78.5±1.1
olmOCR-mix-102570.8 79.3 77.9 45.6 93.7 81.3 78.7 99.3 78.3±1.2
Table 2Finetuning on a single epoch ofolmOCR-mix-0225vsolmOCR-mix-1025, evaluated onolmOCR-
Bench.
3.2 Training
We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned onolmOCR-mix-1025 as
described
in Poznanski et al. (2025). We train for one epoch onolmOCR2-synthmix-1025 using an 8xH100 GPU
node. For each document,
############
```

20. assistant (2026-05-31T04:22:22.272Z)

```
That's Table 2. The column order is confirmed: ArXiv, Old scans math, Tables, Old scans, Headers
& footers, Multi column, Long tiny text, Base, Overall. Now let me find Table 3 (the main
benchmark with the olmOCR 2 row) which appears later/earlier in the doc.
```

21. user (2026-05-31T04:22:26.414Z)

```
=== HEADER BLOCK at 12107 ===
ArXiv
Old
scans
math
Tables Old
scans
Headers
& footers
Multi
column
Long
tiny
text
Base Overall
olmOCR-mix-022578.6 79.9 72.9 43.9 95.1 77.3 81.2 98.9 78.5±1.1
olmOCR-mix-102570.8 79.3 77.9 45.6 93.7 81.3 78.7 99.3 78.3±1.2
Table 2Finetuning on a single epoch ofolmOCR-mix-0225vsolmOCR-mix-1025, evaluated onolmOCR-
Bench.
3.2 Training
We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned onolmOCR-mix-1025 as
described
in Poznanski et al. (2025). We train for one epoch onolmOCR2-synthmix-1025 using an 8xH100 GPU
node. For each document, 28 completions are generated. Each completion gets scored using the
standard
olmOCR-Benchscoring rules, where each test case is either a pass or fail, and the reward is the
fraction
from 0.0 to 1.0 of passing test cases. An example of this reward is shown in Figure 4.
Besides the unit test above, we include two additional rewards to ensure c
================================================
=== HEADER BLOCK at 15975 ===
ArXiv
Old
scans
math
Tables Old
scans
Headers
```

& footers
Multi
column
Long
tiny
text
Base Overall
Mistral OCR API 77.2 67.5 60.6 29.3 93.6 71.3 77.1 99.472.0±1.1
Marker1.10.183.8 66.8 72.9 33.5 86.6 80.0 85.7 99.376.1±1.1
MinerU2.5.4* 76.6 54.6 84.9 33.7 96.6 78.2 83.5 93.775.2±1.1
DeepSeek-OCR 77.2 73.6 80.2 33.3 96.1 66.4 79.4 99.875.7±1.0
Nanonets-OCR2-3B 75.4 46.1 86.8 40.9 32.1 81.9 93.0 99.669.5±1.1
PaddleOCR-VL* 85.7 71.0 84.1 37.8 97.0 79.9 85.7 98.580.0±1.0
Infinity-Parser 7B* 84.4 83.8 85.0 47.9 88.7 84.2 86.4 99.882.5±?
Chandra OCR0.1.0* 82.2 80.3 88.0 50.4 90.8 81.2 92.3 99.983.1±0.9
olmOCR(first release) 63.3 67.5 62.3 38.6 93.4 67.6 54.8 97.968.2±1.1
+Dynamic temp scaling 71.4 73.1 65.6 40.5 93.2 76.6 64.9 96.772.8±1.2
+Better prompting 76.3 76.0 70.2 43.2 94.1 77.5 71.9 96.875.8±1.0
+ New trainer, YAML,
img resize, Qwen 2.5 VL 78.8 77.5 71.9 45.4 94.2 78.6 81.4 99.878.5±1.1
==

22. assistant (2026-05-31T04:22:29.240Z)

This is Table 3 (the main benchmark with baselines). I can see the column order is confirmed and the table continues with ablation rows. Let me get the final olmOCR 2 row that should appear right after the ablation rows shown.

23. user (2026-05-31T04:22:32.242Z)

img resize, Qwen 2.5 VL 78.8 77.5 71.9 45.4 94.2 78.6 81.4 99.878.5±1.1
+Handle blank pages 78.6 79.9 72.9 43.9 95.1 77.3 81.2 98.978.5±1.1
+ Synth data,
RLVR, souping 83.0 82.3 84.9 47.7 96.1 83.7 81.9 99.782.4±1.1
Table 3OCR model performance comparison. Results are reproduced in-house, except those marked with *, which
are reported by model authors.
YAML.The firstolmOCRwas trained to output JSON objects. We switched to YAML, which reduced the retry rate dramatically. We speculate this is because the model does not need to remember how many open
quotes there are currently in the JSON and can simply output an EOS token as soon as it is done.

24. assistant (2026-05-31T04:22:49.902Z)

Definitively confirmed. The final olmOCR 2 row ("+ Synth data, RLVR, souping") is:

`83.0 [ArXiv] 82.3 [Old scans math] 84.9 [Tables] 47.7 [Old scans] 96.1 [Headers&footers] 83.7 [Multi column] 81.9 [Long tiny text] 99.7 [Base] 82.4±1.1 [Overall]`

Mapping each value to the confirmed column header order: **Tables = 84.9**, **Multi column = 83.7**. The WebSearch summary that claimed "84.9 = math formula" was simply wrong (the math/"Old scans math" column is 82.3); it conflated numbers. The primary source is unambiguous.

Now let me assess the claim's full scope against the verified facts:

Numbers: Tables=84.9 ?, Multi-column=83.7 ? ? exact match to claim.

Abstract quote: "the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions" ? verbatim match. The claim's paraphrase ("largest gains specifically to table parsing and multi-column layouts") is accurate, though it omits "math formula conversion." That omission is benign for muneem's framing (it just drops one of the three) and the claim doesn't misrepresent it as the *only* areas.

One nuance to weigh: "largest improvements ... compared to previous versions" is an improvement-delta claim (olmOCR 2 vs olmOCR 1), not a claim that 84.9/83.7 are absolute bests on the benchmark. The data supports this reading: olmOCR-1's tables went 62.3 ? 84.9 (+22.6) and multi-column 67.6 ? 83.7 (+16.1), among the largest per-category deltas. The claim's wording ("attributing its largest gains specifically to table parsing and multi-column") correctly frames it as gains/improvements, not absolute superiority. So this is consistent.

Is olmOCR 2 actually best at tables? No ? and the claim doesn't assert that. Chandra (88.0) and Nanonets-OCR2-3B (86.8) beat 84.9 on tables in the same table. But the claim only says olmOCR 2's *largest gains* were in tables/multi-column, which is true. The claim is narrowly scoped and does not overreach into "best at tables."

The claim is well-supported by the primary source, current (Oct 2025 paper), and the source quality (primary arXiv paper, verified against the actual extracted PDF text) matches the claim strength. The only quibble (dropping "math formula conversion" from the list of three) does not distort the claim for muneem's purposes.

25. user (2026-05-31T04:22:59.974Z)

Structured output provided successfully

26. assistant (2026-05-31T04:23:07.817Z)

Verification complete. The claim is well-supported and not refuted.

I confirmed both numbers directly against the raw PDF text of the primary source (arXiv 2510.19817v1, extracted via pypdf at `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\tool-results\webfetch-1780201298991-vzygw7.pdf`): Table 3's olmOCR 2 row gives Tables = 84.9 and Multi column = 83.7, with the column-header order verified independently. The abstract's "largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions" matches verbatim.

The one initially-conflicting signal (a WebSearch summary claiming 84.9 was the math score) turned out to be a summarization error, refuted by the primary source itself. The claim's only minor liberties ? omitting "math formula conversion" from the trio, and the fact that olmOCR 2 is not the single best table model (Chandra 88.0, Nanonets 86.8 beat it) ? do not contradict what the claim actually asserts (largest *gains*, not absolute best). refuted=false, high confidence.